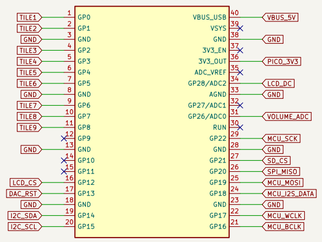
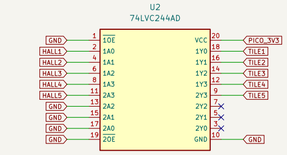
SHAGGY / CARRIER REV A

Pico 2 and existing offboard modules. GPIO assignments preserved.

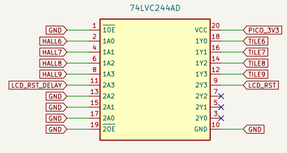
Pico 2 / USB power input



Hall and LCD reset buffers / non-inverting



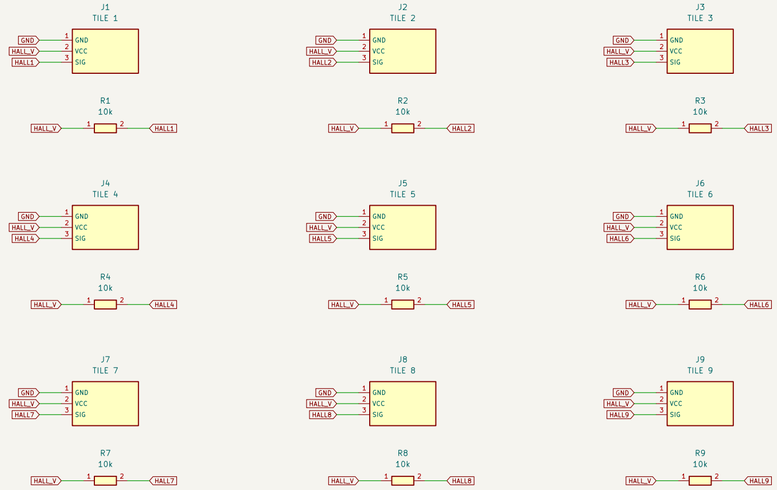




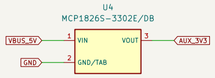


Use Nexperia 74LVC244AD. Inputs tolerate 5 V and have Schmitt action.
Unused inputs are grounded; unused outputs are marked NC.

Nine sensor cables / carrier connector order



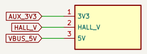
3.3 V auxiliary regulator





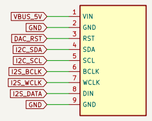






Fit one shunt on JP1. Default 1–2 = 3.3 V.
Check Hall sensor model before selecting 2–3 = 5 V.

Audio cable to Adafruit TLV320DAC3100







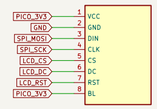






Speaker remains on the DAC SPK+ and SPK− outputs.
Neither speaker wire connects to carrier GND.
DAC I2C pullups are on the Adafruit module.

Waveshare 1.69 inch LCD





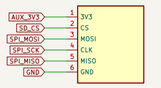






RST = 10k / 100nF delay through U3 Schmitt buffer.
R10 = 100k for the LCD TXB010B. BL stays on.
Validate buffered power-on reset on cold starts and fast cycles.

SPI microSD reader

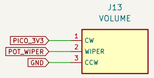








Volume potentiometer





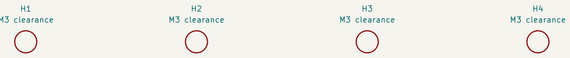


SPI source termination





Mounting / M3 clearance



Carrier cable order is printed here. Trace each module's pin labels before connecting.
Hall tile positions are firmware order: trace the physical 3 × 3 panel before labeling cables.
Pico USB is the only power input. Measure total current with all voices and the display active.
PICO_3V3 powers the LCD, potentiometer and buffers. AUX_3V3 powers SD and the selected Hall rail.
U4 dissipates 0.51 W at 5 V input / 300 mA output. Verify regulator temperature inside the enclosure.
The 100 × 80 mm outline and mounting holes are provisional. Enclosure dimensions are unmeasured.

100 × 80 mm provisional carrier. Verify fit before fabrication.

Shaggy

Sheet: /
File: shaggy-carrier.kicad_sch

Title: Shaggy breadboard replacement

Size: A2	Date: 2026-10-29	**Rev: A — prototype**
KiCad E.D.A. 10.0.7		Id: 1/1